

# KENNY CAO

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## Education

### University of California, San Diego

Expected June 2029

*Bachelor of Science in Computer Science, Emphasis in Mathematics*

*La Jolla, CA*

– **Relevant Coursework:** Data Structures & Algorithms, Databases, Operating Systems

## Experience

### Stealth Startup

July 2025 – Present

*Software Engineering Intern*

- Engineered an agentic full-stack sales automation platform that leverages autonomous LLM agents to qualify leads and orchestrate B2B outreach workflows.
- Built a high-concurrency backend using FastAPI and LangChain, optimizing agent tool-calling efficiency by implementing a Redis-based state management system.
- Improved lead conversion rates by 35% and reduced manual data entry overhead for sales teams by 50+ hours per week through automated CRM synchronization.

### University of California, San Diego

Sept. 2025 – Present

*Software Research Assistant*

*La Jolla, CA*

- Developed an LLM-powered narrative engine designed to generate state-aware, dynamic quest lines and NPC interactions in interactive adventure games.
- Optimized context window utilization by building a custom vector-retrieval system (RAG) in Python to inject relevant world-building data into model prompts.
- Achieved a 20% increase in response formatting accuracy and reduced token consumption by 15% through refined few-shot prompt engineering.

## Projects

### Distributed Real-Time Trading Engine | *Python, Redis, Docker, AWS*

January 2026

- Architected a high-frequency trading simulation engine capable of processing 10,000+ mock transactions per second with sub-millisecond latency.
- Deployed a containerized microservices architecture using Docker and AWS ECS, utilizing Redis Pub/Sub for low-latency communication between services.

### Omni-Auth: Secure Microservices Gateway | *Go, gRPC, JWT, PostgreSQL*

Oct. 2025

- Engineered a distributed authentication gateway for microservices to centralize identity management and session validation across a heterogeneous tech stack.
- Implemented JWT-based stateless authentication using gRPC interceptors and optimized session lookups by utilizing a write-through caching strategy.
- Reduced cross-service authentication latency by 25% and centralized security policy enforcement for over 15 distinct API endpoints.

### Scale-Share: Cloud-Native Image Host | *Go, AWS S3, CloudFront, Terraform*

Aug. 2025

- Designed a scalable image hosting service designed to handle global content delivery with high availability and automated optimization.
- Provisioned infrastructure as code using Terraform to manage AWS S3 buckets and CloudFront distributions, ensuring 99.9% uptime for global asset delivery.

### Dynamic Desktop Utility Suite | *JavaScript, Figma, GitHub*

June 2024 – Present

- Developed and published front-end applications on the Steam platform, focusing on user-interactive utility software and custom UI frameworks.
- Optimized UI/UX responsiveness and integrated telemetry features to track user engagement and software performance.

## Leadership

### Association for Computing Machinery

August 2025 – Present

*Software Engineering Lead*

*La Jolla, CA*

- Taught fifty students full-stack development and industry-standard practices through weekly technical workshops.
- Organized campus-wide hackathons resulting in a thirty percent increase in student participation.

### UCSD Recreation Center

Fall 2025 – Present

*Student Leader*

*La Jolla, CA*

- Directed strategic planning and tournament timelines for campus-wide competitive esports initiatives.
- Analyzed performance metrics to develop data-driven team growth goals and training regimens.

## Technical Skills

**Languages:** Python, Java, JavaScript, Go, HTML, XML, Lua, SQL

**Tools & Frameworks:** Node.js, React.js, FastAPI, Git, Figma, PyTorch, Redis, Docker, AWS (S3, ECS)

**Developer Tools:** VS Code, GitHub, Terraform, Android Studio, Linux